

Sachin Bala

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Technical Skills

Six Sigma (Yellow Belt), Lean Manufacturing (5S, Kaizen), PFMEA, DFMEA, FMEA, DOE, SPC, Process Capability Studies, Root Cause Analysis (8D, 5-Why), Python (Pandas, NumPy, scikit-learn), SQL (MySQL), R, JMP, Minitab, Data Visualization (Grafana, Matplotlib), Machine Learning, Computer Vision (YOLO), LabVIEW, SolidWorks, AutoCAD, Cognex, Dr Schenk vision system, Microsoft Suite

Education

Western Michigan University (WMU)

Overall GPA: 3.69

Bachelor's in Mechanical Engineering with minor in Computer Science and Mathematics

Work Experiences

Associate Process Development Engineer, Tesla Inc., Austin, TX

02/25-present

- Translated manufacturing operational needs into structured process requirements and qualification protocols; commissioned two electrode production lines achieving 98% yield through end-to-end production, quality, and performance validation.
- Led operational excellence initiatives applying Lean and Six Sigma methodologies (FMEA, DOE, SPC, 8D, 5-Why, Capability Study) to identify root causes of yield excursions and drive measurable improvements across production, quality, maintenance, and inventory processes.
- Developed data-driven dashboards and analytics pipelines using MySQL, Python, R, and JMP to surface real-time operational insights; enabled data-backed investment and process decisions across the manufacturing organization through cross-functional stakeholder collaboration.

Engineering Technician, Process Development Team, Tesla Inc., Austin, TX

8/23-02/25

- Digitized shop floor monitoring by designing and deploying a real-time Grafana dashboard tracking tool performance metrics across the production floor; authored SOPs for sensor integration, enabling consistent and reliable operational data capture.
- Supported tool qualification and process validation against product specifications using statistical capability studies; identified and implemented process improvements that increased production rate by 10% while sustaining quality targets.
- Conducted structured root cause analysis on downtime events using SPC charts, fault logs, and process historians; implemented corrective actions that reduced unplanned stoppages and improved operational uptime -- directly supporting shop floor non-conformance management and change management activities.

Engineering Co-Op, Scherdel SST, Muskegon, MI

8/22-12/22

- Researched EV component manufacturing material trends and synthesized findings to support product launch decisions and quality process design; delivered quantitative insights to cross-functional stakeholders across engineering and operations.

Process Engineering Intern, Scherdel SST, Muskegon, MI

05/22-08/22

- Analyzed spring manufacturing process end-to-end to detect early failure modes; built a predictive model using ML algorithms and statistical analysis, improving production efficiency and accuracy by 25%. Validated model using Gauge R&R and F1-score methodology.
- Developed a Computer Vision defect detection system to automate quality inspection on the production line; validated system reliability using structured quantitative methods (Gauge R&R, F1-score analysis).

Laser Tech Technician Internship, Technique Inc., Jackson, MI

09/21-04/22

- Optimized laser cutting processes using CAD and machine troubleshooting; validated chassis parts via design review and laser tracker measurement data.
- Updated SOPs and trained technicians and operators to follow standardized procedures, supporting change management and consistent shop floor execution.

Engineering Research and Project

Senior Design Project - Two Tier Industrial Cart (Technique Inc.)

01/22-12/22

- Designed a two-tier industrial cart in SolidWorks with scissor lift mechanism; validated with FEA analysis and

numerical calculations.

- Applied DFMEA to identify and mitigate risk, establishing data-backed design controls and a quality control plan.

Honors and Membership

Diether H. Haenicke Scholarship, WMU Dean's List, Yellow Belt Training (Six Sigma), Root Nepal (Co-Founder)